

Name: _____ Date: _____

Period: _____

1. For the rational function $r(x) = \frac{5}{(x-3)(x+1)}$, find all vertical asymptotes. Write each as an equation.

Vertical asymptote(s): _____

2. A table of values near $x = 2$ is shown below.

x	1.9	1.99	2.001	2.01	2.1
$r(x)$	47	497	501	51	5.7

Based on the table, what does the behavior of $r(x)$ as $x \rightarrow 2$ indicate about $x = 2$? Circle the best answer and explain briefly.

A. Hole at $x = 2$ **B.** Vertical asymptote at $x = 2$ **C.** Neither

Explanation: _____

3. Factor the expression $x^2 + x - 6$ completely.

 $x^2 + x - 6 =$ _____